

Ch2 Tangent Vectors and Differentials

Germes of Functions & The Tangent Space

LXJ

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The Idea: Derivations

Tangent vectors as operators

Idea. Any vector $v = (v_1, \dots, v_n) \in \mathbb{R}^n$ can be thought of as an *operator* on differentiable functions. If f is differentiable at p , then v assigns to f the *directional derivative*

$$v(f) = \sum_{i=1}^n v_i \frac{\partial f}{\partial r_i}(p), \quad \text{which is a real value.}$$

It satisfies

1. **linearity** $v(\lambda f + \mu g) = \lambda v(f) + \mu v(g);$
2. **Leibniz rule** $v(f \cdot g) = f(p) \cdot v(g) + v(f) \cdot g(p).$

Why germs of functions?

The operation of taking derivatives only depends on the *local* properties of functions. This is captured by the concept of *germs of functions*.

Remark

A vector v is thus “defined” as a linear function on differentiable functions, determined by the values of $v(f)$.

Definition (Germs of functions)

Let $m \in M$. Functions f and g defined on open sets containing m are said to have the *same germ at m* if they agree on some neighborhood of m . The equivalence classes of germs of C^∞ functions are called *germs*, denoted by $\tilde{\mathcal{F}}_m$, i.e.

$$\tilde{\mathcal{F}}_m = \{[f] \mid f, g \in [f] \text{ if } \exists \text{ open set } U \ni m, \text{ s.t. } f|_U = g|_U\}.$$

The Algebra of Germs

Operations on $\tilde{\mathcal{F}}_m$

The addition, scalar multiplication and multiplication of functions induce

$$(\text{addition}) \quad [f] + [g] = [f + g],$$

$$(\text{scalar mult.}) \quad \lambda[f] = [\lambda f],$$

$$(\text{multiplication}) \quad [f] \cdot [g] = [fg].$$

Exercise

Prove that these are well-defined operations.

Remark

These operations make $\tilde{\mathcal{F}}_m$ an *algebra* over $\mathbb{R}$. A germ $[f]$ has a well-defined value $f(m)$ at m .

Definition (The ideals F_m and F_m^k)

$$F_m = \{[f] \mid f \in \tilde{\mathcal{F}}_m, f(m) = 0\}.$$

Powers of the ideal F_m

F_m is an *ideal* of $\tilde{\mathcal{F}}_m$. Let F_m^k be the k -th power of F_m : the ideal of $\tilde{\mathcal{F}}_m$ consisting of all finite linear combinations of k -fold products of elements of F_m ,

$$F_m^k = \{\text{linear comb. of } [f_1] \cdots [f_k], [f_i] \in F_m\}.$$

These form a descending sequence of ideals

$$\tilde{\mathcal{F}}_m \supset F_m \supset F_m^2 \supset F_m^3 \supset \cdots$$

The Tangent Space

Tangent vectors as derivations

Definition (Tangent vector)

A *tangent vector* v at the point $m \in M$ is a *linear derivation* of the algebra $\tilde{\mathcal{F}}_m$, i.e. for $[f], [g] \in \tilde{\mathcal{F}}_m$ and $\lambda, \mu \in \mathbb{R}$:

$$(a) \quad v(\lambda[f] + \mu[g]) = \lambda v([f]) + \mu v([g]);$$

$$(b) \quad v([f] \cdot [g]) = v([f]) g(m) + f(m) v([g]).$$

Let $T_m M$ or M_m denote the set of all linear derivations at m , called the *tangent space to M at m* .

Note

If $M_m \neq \emptyset$, then M_m is a *linear space*, since for $v, w \in M_m$, $\lambda, \mu \in \mathbb{R}$,

$$(\lambda v + \mu w)([f]) \triangleq \lambda v([f]) + \mu w([f]).$$

Two remarks on the definition

1. This definition of tangent spaces only works for the C^∞ case;
2. It is *intrinsic* — it does not depend on the local coordinate system.

Note (constants are killed)

For a derivation $w \in M_m$ and $[c]$ the germ of a locally constant function, $w([c]) = 0$.

Proof.

It suffices to show $w([1]) = 0$:

$$w([1]) = w([1] \cdot [1]) = w([1]) \cdot 1 + 1 \cdot w([1]) = 2 w([1]).$$

□

$$M_m \cong (F_m/F_m^2)^*$$

The key isomorphism

Lemma

M_m is isomorphic to $(F_m/F_m^2)^*$.

Proof.

Consider $v \in M_m$. Then v is a linear function on F_m by $v : [f] \mapsto v([f])$. This function *vanishes on* F_m^2 by the Leibniz rule of v : $\forall [f], [g] \in F_m$,

$$v([f][g]) = v([f]) \cdot g(m) + f(m) \cdot v([g]) = 0.$$

So v can be thought of as a linear function defined on the cosets $[f] + F_m^2$, therefore $v \in (F_m/F_m^2)^*$. □

Converse direction

Conversely, for $\ell \in (F_m/F_m^2)^*$, we can define $v_\ell \in M_m$ by

$$v_\ell([f]) \triangleq \ell([f] - [f(m)] + F_m^2).$$

Then one verifies that v_ℓ is a derivation on $\tilde{\mathcal{F}}_m$:

1. Linearity.

$$v_\ell(\lambda[f] + \mu[g]) = \ell(\lambda[f] - \lambda[f(m)] + F_m^2) + \ell(\mu[g] - \mu[g(m)] + F_m^2) = \lambda v_\ell([f]) + \mu v_\ell([g]).$$

2. Leibniz. $v_\ell([f][g]) = \ell([f][g] - [f(m)][g(m)] + F_m^2)$, where

$$\begin{aligned} [f][g] - [f(m)][g(m)] &= \underbrace{([f] - [f(m)])([g] - [g(m)])}_{\in F_m^2} \\ &\quad + [f(m)]([g] - [g(m)]) + ([f] - [f(m)])[g(m)], \end{aligned}$$

$$\text{so } v_\ell([f][g]) = f(m) v_\ell([g]) + v_\ell([f]) g(m).$$

An exercise hidden in the proof

Exercise

Prove that for $\ell \in (F_m/F_m^2)^*$,

$$\ell([f(m)] \cdot ([g] - [g(m)])) = f(m) \ell([g] - [g(m)]).$$

Theorem

$$\dim(F_m/F_m^2) = \dim M.$$

Proof of $\dim(F_m/F_m^2) = \dim M$

A Taylor lemma with integral remainder

Lemma

If $g \in C^k(U)$ for $k \geq 2$ and U is a convex open set in $\mathbb{R}^d$, then $\forall p, q \in U$,

$$\begin{aligned} g(q) = g(p) &+ \sum_{i=1}^d \left. \frac{\partial g}{\partial r_i} \right|_p (r_i(q) - r_i(p)) \\ &+ \sum_{i,j} (r_i(q) - r_i(p)) (r_j(q) - r_j(p)) \int_0^1 (1-t) \left. \frac{\partial^2 g}{\partial r_j \partial r_i} \right|_{p+t(q-p)} dt. \end{aligned}$$

Proof.

Since U is convex, for any $p, q \in U$ the segment $\{p + t(q - p) \mid t \in [0, 1]\} \subset U$. Writing $p_i \triangleq r_i(p)$, $q_i \triangleq r_i(q)$, we get

$$g(q) - g(p) = \int_0^1 \frac{d}{dt} g(p + t(q - p)) dt = \sum_i \int_0^1 \left. \frac{\partial g}{\partial r_i} \right|_{p+t(q-p)} (q_i - p_i) dt. \quad \square$$

Proof of the Taylor lemma (continued)

For each integrand, expand around p :

$$\left. \frac{\partial g}{\partial r_i} \right|_{p+t(q-p)} = \left. \frac{\partial g}{\partial r_i} \right|_p + \sum_j \int_0^t \frac{\partial^2 g}{\partial r_j \partial r_i} (p + s(q-p)) (q_j - p_j) ds.$$

So, interchanging the order of integration ($s \leftrightarrow t$),

$$\begin{aligned} g(q) = g(p) &+ \sum_i \left. \frac{\partial g}{\partial r_i} \right|_p (q_i - p_i) \\ &+ \sum_{i,j} (q_i - p_i)(q_j - p_j) \int_0^1 (1-t) \left. \frac{\partial^2 g}{\partial r_j \partial r_i} \right|_{p+t(q-p)} dt. \end{aligned}$$

Proof of the theorem: spanning set

Let (U, φ) be any coordinate system about $m \in M$, with coordinate functions $(x_1, \dots, x_d)$, $d = \dim M$. Let $[f] \in \tilde{\mathcal{F}}_m$. Applying $f \circ \varphi^{-1}$ to the lemma:

$$\begin{aligned} f \circ \varphi^{-1}(x) - f \circ \varphi^{-1}(x_0) &= \sum_{i=1}^d \left. \frac{\partial(f \circ \varphi^{-1})}{\partial x_i} \right|_{x_0} (x_i - x_i^0) \\ &\quad + \sum_{i,j} (x_i - x_i^0)(x_j - x_j^0) \int_0^1 (1-t) \left. \frac{\partial^2(f \circ \varphi^{-1})}{\partial x_j \partial x_i} \right|_{x_0 + t(x - x_0)} dt. \end{aligned}$$

Since $x = \varphi(p)$, $x_0 = \varphi(m)$, composing with φ gives, *modulo* F_m^2 ,

$$[f(p)] - [f(m)] = \sum_{i=1}^d \left. \frac{\partial(f \circ \varphi^{-1})}{\partial x_i} \right|_{\varphi(m)} (x_i(p) - x_i(m)).$$

Hence $\{[x_i] - [x_i(m)]\}$ spans F_m/F_m^2 .

Proof of the theorem: linear independence

We claim they are also *linearly independent*. Suppose that

$$\sum_i a_i(x_i - x_i(m)) \in F_m^2 \iff \sum_i a_i(r_i - r_i(\varphi(m))) \in F_{\varphi(m)}^2.$$

Then applying $\left. \frac{\partial}{\partial r_j} \right|_{\varphi(m)}$ to both sides gives

$$a_j = 0 \quad \forall j.$$

Therefore $\{[x_i] - [x_i(m)]\}_{i=1}^d$ is a basis of F_m/F_m^2 , and

$$\dim(F_m/F_m^2) = d = \dim M. \quad \square$$